CHAPTER 14

CALCULATING BILLING DATA

In this chapter you'll learn about

- Preparing a Billing with Data from dBASE or Lotus
- Math Capabilities at Print Time
- Variable Number and Text Formatting
- Using Conditional Statements and Paragraphs

Preparing a billing involves combining text and data. You need the customer's name and address and the amount due. You may also need other specialized information, such as which regional office the customer should contact if he or she has any questions. You may also have individualized notes to add to certain customers' letters.

WordStar imports information from many database programs. You can add data about a particular customer without editing the file each time. You can even have WordStar figure the total due when you print the document.

The steps in this chapter specify using a dBASE III file. To follow the instructions exactly, you'll need a dBASE III file containing customers' names, addresses, and account data. If you use another program, adapt the instructions as necessary.

Note: If you're using a dBASE IV file with memo fields, use dBASE to convert it to a dBASE III file. WordStar handles the dBASE III format more easily.

CREATING A BILLING DOCUMENT

In the following steps, you'll create a form letter to serve as a billing. WordStar fills in the blanks at print time to create an individual billing for each customer.

1. Open a new file and name it BILLING.DOC.

2. Choose Utilities→Merge Print Commands→Data File (Alt+U,P,D). Type the name of the dBASE file containing your customer information. Include the complete path to the file. Press F10 or choose OK.

   WordStar inserts the dot command **.df** into the file.

3. Choose Utilities→Merge Print Commands→Name Variables (Alt+U,P,N). In the Variables box type * to use all of the predefined field names in your dBASE file as WordStar variables. If you prefer, you can type the variable names yourself in the same order as they appear in the data file. Press F10 or choose OK.

   WordStar inserts the dot command **.rv** into the file.

4. Choose Layout→Page Numbering (Alt+L,N) to display the Page Numbering dialog box. Clear the Print Page Numbers check box. This turns off page numbering. Press F10 or choose OK.

   WordStar inserts the dot command **.op** into the file.

TYPING TEXT AND VARIABLES

In the next steps, you type the text and variables for your billing.

Ampersands (&) around text indicate a variable. Variables are replaced by actual data at print time. You add the /o option to the customer address fields to ensure that lines won't be left blank if some customers don't have a second address line.

The &@& date variable is a system variable—the information for that variable is inserted from your computer's memory. If the date device in your computer is set correctly, the correct date replaces this variable when the billing is printed. The date needs more space than the variable does, so remember to leave enough room.

1 Type the text for your billing using the following example. If necessary, use your dBASE variable names instead of those shown in the example.

```
&Mr-Ms& &full-name&
&addr1/o&
&addr2/o&
&city&, &state& &zip&                                    &@&

Dear &Mr-Ms& &full-name&:

Per your last order:
&units1& &equip1& at &costper1& =
&units2& &equip2& at &costper2& =
&units3& &equip3& at &costper3& =
                     Total due

Sincerely,

Mr. Baxter
Baxter and Associates
```

2 Choose Insert→Page Break (Alt+I,P) to insert a page break.

WordStar inserts the dot command **.pa** at the end of the text.

DETERMINING TOTALS AT PRINT TIME

Now that you've typed the text for your billing, you can use the WordStar math features to calculate the subtotals and the grand total due. When you print the file, WordStar makes these calculations for each billing.

CALCULATING THE SUBTOTALS

1. Move the cursor to the **.op** command. The cursor should be in column 1.

2. Choose Utilities→Merge Print Commands→Set Variable to Math Result (Alt+U,P,V). In the Variable Name box, type **total1**. In the Mathematical Expression to be Calculated box, type **&units1& * &costper1&**. Press F10 or choose OK.

   WordStar inserts the following command:

   ```
   .ma total1 = &units1& * &costper1&
   ```

   The **.ma** command determines the subtotal.

3. Choose Utilities→Merge Print Commands→Set Variable to Math Result (Alt+U,P,V). In the Variable Name box, type **total2**. In the Mathematical Expression to be Calculated box, type **&units2& * &costper2&**. Press F10 or choose OK.

4. Choose Utilities→Merge Print Commands→Set Variable to Math Result (Alt+U,P,V). In the Variable Name box, type **total3**. In the Mathematical Expression to be Calculated box, type **&units3& * &costper3&**. Press F10 or choose OK.

   The **.ma** commands should look like this:

   ```
   .ma total1 = &units1& * &costper1&
   .ma total2 = &units2& * &costper2&
   .ma total3 = &units3& * &costper3&
   ```

   The **.ma** command creates new variables, total1, total2, and total3, which represent the subtotals for each item ordered.

5. Move the cursor to the space following the first equal sign (=) in the text of the letter.

6. Press the Spacebar and type **&total1&**.

7. Move to the space following the next equal sign (=) , press the Spacebar and type **&total2&**. Do the same for the third subtotal and type **&total3&**.

CALCULATING THE GRAND TOTAL

You can add another **.ma** command to find the grand total due. WordStar prints this amount on the line that represents the total due.

1. Move the cursor to the **.op** command.

2. Choose Utilities→Merge Print Commands→Set Variable to Math Result (Alt+U,P,V). In the Variable Name box, type **gtotal**. In the Mathematical Expression box, type **&total1& + &total2& + &total3&**. Press F10.

   The following command appears in your file:

   ```
   .ma gtotal = &total1& + &total2& + &total3&
   ```

3. Move the cursor to the space following the words "Total due."

4. Press the Spacebar and type **&gtotal&**.

   Your text should look like this:

   ```
   Total due &gtotal&
   ```

VARIABLE NUMBER FORMATTING

Now you'll add some commands to tell WordStar how to print the numbers to be inserted at print time.

1. Move the cursor to the **.op** command.

2. Choose Utilities→Merge Print Commands→Set Variable (Alt+U,P,S). In the Variable box, type **1**. In the Data box, type **($$$,$$$.99)**. Press F10 or choose OK.

   The following command appears in your file:

   ```
   .sv 1=($$$,$$$.99)
   ```

This format tells WordStar to put a dollar sign in front of variables with the /1 parameter (see step 3 below). If the number is more than 999, a comma is inserted after the thousands place. If the number is less than 1,000, the dollar sign is preceded by spaces. If the number is less than zero (for instance, if the customer has a credit), the number is in parentheses.

3 Add the /1 parameter to each variable in the text that represents a dollar amount. In your file, these are the three &costper& variables, three &total& variables, and the &gtotal& variable. For example, the &total1& variable now looks like this:

```
&total1/1&
```

VARIABLE TEXT FORMATTING

You can control the format of variables that print as text. In the following steps you'll left-align the &equip& variables you typed in the text.

1 In your file, move the cursor to the **.op** command

2 Choose Utilities→Merge Print Commands→Set Variable (Alt+U,P,S). In the Variable box, type **2**. In the Data box, type **L** 15 times. Press F10 or choose OK.

The following command appears in your file:

```
.sv 2=LLLLLLLLLLLLLL.
```
(There are 15 L's.)

This command tells WordStar to allow 15 spaces for the name of each kind of equipment purchased. The equipment name aligns at the left margin. If the equipment name is less than 15 characters long, WordStar adds as many spaces as necessary to fill the 15-character space.

3 Move the cursor to the &equip& variables in your text. Add the **/2** parameter to each one. The &equip1& variable now looks like this:

```
&equip1/2&
```

USING CONDITIONAL STATEMENTS

Sometimes the information you send to customers varies, depending on where they live, the amount of their balance, or the date of the billing. In the following steps, you'll add a paragraph called a *conditional paragraph*, that tells customers which regional branch of your company to contact, based on the customer's address.

Follow these steps to add a conditional paragraph:

1. Move the cursor to the blank line above the word "Sincerely."

2. Choose Utilities→Merge Print Commands→If (Alt+U,P,I). In the Variable Name box, type **&state&**. In the Constant box, type **CA**. From the Operator (Alphabetical) options, select = same as. Press F10 or choose OK.

   The following command appears in your file:

   ```
   .if &state&=  CA
   ```

   The **.if** command tells WordStar to use the Western Regional Office paragraph (see step 3) for customers living in CA (California).

3. Press Enter after the **.if** command and type the paragraph containing the Western Regional Office. For example, type

   ```
   If you have questions about your billing, contact Gabriel
   Grace at our Western Regional Office, 12 San Pablo Road,
   San Rafael, CA 94903.
   ```

4. Move the cursor to the next line. Choose Utilities→Merge Print Commands→Else (Alt+U,P,L) to specify the paragraph to print if the customer lives in a state other than California.

   The **.el** command appears in your file.

5. Type the paragraph to use for the other customers. For example, type

```
If you have questions about your billing, contact Morgan
J. Robinson at our Main Office, 1201 27th Street,
Rockaway, NY 11691.
```

6 On the next line, choose Utilities→Merge Print Commands→End If (Alt+U,P,E) to end the condition.

Now, every customer in California will receive a billing containing the Western Regional Office address. Other customers will receive a billing containing the Main Office address.

CONDITIONAL PARAGRAPHS BASED ON THE BALANCE DUE

You may want your billing to include an announcement of a special offer or sale. In the following example, you'll add a paragraph announcing a discount for customers. The size of the discount will depend on the balance due.

Customers who owe the most receive the biggest discount. You'll add a conditional statement offering a 15% discount to customers whose balance is more than $300.00.

1 Move the cursor to the line just above "Sincerely," (create a blank line if necessary).

2 Choose Utilities→Merge Print Commands→If (Alt+U,P,I). In the Variable Name box, type **&gtotal&**. In the Constant box, type **300.00**. From the Operator (Numerical) options, select # greater than. Press F10 or choose OK.

This command appears in the file:

```
.if &gtotal& #> 300.00
```

3 Type the paragraph offering the 15% discount. For example, on the next line type

```
Because you're such a good customer, we're offering a
special 15% discount on all merchandise for the next
month.  Take advantage of this special offer to stock up
on all your equipment needs.
```

Now tell WordStar to offer a 10% discount to the customers whose balance is less than $300.00 but more than $200.00.

1. Choose Utilities→Merge Print Commands→Else (Alt+U,P,L).

2. Choose Utilities→Merge Print Commands→If (Alt+U,P,I). In the Variable Name box, type **&gtotal&**. In the Constant box, type **200.00**. From the Operator (Numerical) options, select # greater than. Press F10 or choose OK.

3. Type the paragraph offering the 10% discount.

```
We're grateful for your patronage, and we're offering you
a special 10% discount on all merchandise for the next
month.  Take advantage of this special offer to stock up
on all your equipment needs.
```

Now you'll type a paragraph for customers who don't earn a discount.

1. Choose Utilities→Merge Print Commands→Else (Alt+U,P,L). This tells WordStar to print the next paragraph if the previous conditions aren't met.

   WordStar inserts the **.el** command.

2. On the line following the **.el** command, type **Thank you for shopping with us.** and press Enter.

Now you'll end the conditions by inserting two **.ei** commands.

1. Choose Utilities→Merge Print Commands→End If (Alt+U,P,E).

   WordStar inserts the first **.ei** command.

2. Repeat step 1 to insert another **.ei** dot command.

SKIPPING RECORDS IN THE DATA FILE

You don't want to bill customers who have no balance due. You can avoid printing a bill for these customers by using the following technique.

1. Move the cursor to the **.op** command.

2. Choose Utilities→Merge Print Commands→If (Alt+U,P,I). In the Variable Name box, type **&gtotal&**. In the Constant box, type **0**. From the Operator (Numerical) options, select #> greater than. Press F10 or choose OK.

3. Choose Edit→Go to Other→Document End (Alt+E,O,O) to move the cursor to the end of the document (after the page break).

4. Choose Utilities→Merge Print Commands→Else (Alt+U,P,L).

   WordStar inserts the **.el** command into the file.

5. On the next line, choose Utilities→Merge Print Commands→Display Message (Alt+U,P,M). In the Message box, type **&full-name& has no balance due. No billing will be printed.** Press F10 or choose OK.

   WordStar inserts a **.dm** command followed by the message you just typed.

   When WordStar finds a customer who has no balance due, it displays this message onscreen and doesn't print a billing.

6. Choose Utilities→Merge Print Commands→End If (Alt+U,P,E) to end the conditional statement.

INCLUDING DBASE III MEMO FIELDS

The dBASE III and dBASE III Plus programs have a memo field feature that allows you to make extensive notes about each record in your data file. If you want to include the dBASE notes for each customer, follow these steps:

1. Move the cursor to the place in your billing where you want WordStar to print the memo field.

   In this example, the memo field is called NOTE, so the variable for that field is &note& (the variable name must always match the field name).

2. Type **&note&** in your file. The entire memo field will print at the &note& variable.

THE BILLING DOCUMENT

Now that you've completed the steps in this chapter, your billing document looks like this:

```
.df c:\dbase\customers.dbf
.rv*
.ma total1=&units1& * &costper1&
.ma total2=&units2& * &costper2&
.ma total3=&units3& * &costper3&
.ma gtotal=&total1& + &total2& + &total3&
.sv 1=($$$,$$$.99)
.sv 2=LLLLLLLLLLLLLL
.if &gtotal& #> 0
.op
&Mr-Ms& &full-name&
&addr1/0&
&addr2/0&
&city&, &state& &zip&                                        &@&

Dear &Mr-Ms& &full-name&:

Per your last order:
&units1& &equip1/2& at &costper1/1& = &total1/1&
&units2& &equip2/2& at &costper2/1& = &total2/1&
&units3& &equip3/2& at &costper3/1& = &total3/1&
                         Total due     &gtotal/1&
.if &state&=CA
If you have questions about your billing, contact Gabriel
Grace at our Western Regional Office, 12 San Pablo Road,
San Rafael, CA, 94903.
.el
```

```
If you have questions about your billing, contact Morgan
J. Robinson at our Main Office, 1201 27th Street,
Rockaway, NY 11691.
.ei
.if &gtotal& #> 300.00
Because you're such a good customer, we're offering a
special 15% discount on all merchandise for the next
month.  Take advantage of this special offer to stock up
on all your equipment needs.
.el
.if &gtotal& #> 200.00
We're grateful for your patronage, and we're offering you
a special 10% discount on all merchandise for the next
month.  Take advantage of this special offer to stock up
on all your equipment needs.
.el
Thank you for shopping with us.
.ei
.ei
Sincerely,

Mr. Baxter
Baxter and Associates
&note&
.pa
.el
.dm &full-name& has no balance due.  No billing will be
printed.
.ei
```

When you save and print the file, a copy of the billing is printed for each customer with a balance due. If the customer lives in California, the paragraph about the regional office is printed. Otherwise, the paragraph about the main office is printed. If the customer has a balance due of $300.00 or more, a 15% discount is offered. If the customer has a balance between $200.00 and $300.00, a 10% discount is offered. If no balance is due, no billing is printed, and you see a message onscreen to that effect.